Using Deep Learning and Satellite Imagery to Detect Forest Areas and Simulate Deforestation

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Abstract

Deforestation is a major environmental challenge that affects ecosystems, biodiversity, and climate stability. Monitoring forest changes across large areas is difficult using traditional methods alone. This project investigates how artificial intelligence and satellite imagery can be used to classify land-cover types and identify forest areas.

A Convolutional Neural Network (CNN) was developed using the EuroSAT satellite image dataset, which contains 16,200 Sentinel-2 satellite images across 10 land-cover categories. The model was trained to recognize patterns in satellite images and classify them into different land types, including forests, agricultural areas, water bodies, and urban regions.

The developed model achieved a training accuracy of 90.04% and a test accuracy of 91.84%. The AI system successfully identified forest areas and estimated that 11.01% of the analyzed images represented forest land. A deforestation simulation was also performed by reducing forest-classified images by approximately 20%, demonstrating how AI-based systems could be used to analyze potential forest loss.

The results show that deep learning models can effectively analyze satellite imagery and support environmental monitoring. Future improvements could include using real satellite images from multiple years to detect actual deforestation over time and creating automated forest-change maps.

Keywords: Artificial Intelligence, Deep Learning, Satellite Imagery, Convolutional Neural Network, Deforestation Detection, Forest Monitoring

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1. Introduction

Deforestation is one of the major environmental challenges affecting ecosystems around the world. Forests provide habitats for wildlife, help regulate climate, and maintain the balance of natural ecosystems. However, forests are continuously affected by activities such as agriculture, urban expansion, and resource extraction.

Monitoring forests over large areas can be difficult using traditional methods alone. Satellite imagery provides a powerful solution by allowing scientists to observe Earth's surface and collect information about different land-cover types. However, analyzing thousands of satellite images manually is time-consuming.

Artificial intelligence (AI) and machine learning can help solve this problem by automatically analyzing satellite images. In this project, a Convolutional Neural Network (CNN) was developed to classify satellite images into different land-cover categories and identify forest areas. The project also demonstrates how AI-based classification can be used to simulate potential deforestation changes.

2. Research Question

How accurately can a convolutional neural network classify satellite images and identify forest areas for monitoring possible deforestation?

3. Project Objective

The objective of this project is to develop an artificial intelligence model that can analyze satellite imagery, classify different land-cover types, detect forest areas, and estimate changes in forest coverage.

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4. Dataset

This project uses the **EuroSAT dataset**, which contains satellite images collected from the **Sentinel-2 satellite mission**.

The dataset includes:

- **16,200 satellite images**
- **13 spectral bands**
- **10 different land-cover classes**

The land-cover categories include:

1. Annual Crop
2. Forest
3. Herbaceous Vegetation
4. Highway
5. Industrial
6. Pasture
7. Permanent Crop
8. Residential
9. River
10. Sea/Lake

The use of multiple spectral bands allows the model to analyze information beyond normal visible colors, improving satellite image classification.

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5. Methodology

Data Preparation

The satellite images were loaded into a Python environment using Google Colab. The dataset was prepared using PyTorch tools and converted into a format suitable for deep learning.

The images were organized into batches and provided to the neural network during training.

Machine Learning Model

A **Convolutional Neural Network (CNN)** was created to classify satellite images.

A CNN was selected because it is effective at recognizing patterns in images. The model learns important features such as:

- Vegetation patterns
- Water regions
- Buildings
- Agricultural areas
- Land textures

The model architecture included:

- Convolutional layers for feature extraction
- ReLU activation functions
- Pooling layers to reduce image size
- Fully connected layers for classification

The model was trained using:

- Optimizer: Adam
- Loss Function: Cross Entropy Loss
- Training Epochs: 10

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6. Model Training Results

The CNN model successfully learned patterns from satellite imagery.

Results:

Training Accuracy: 91%

```
Epoch 1/10 Loss: 226.7488 Accuracy: 84.86%
Epoch 2/10 Loss: 223.2614 Accuracy: 85.16%
Epoch 3/10 Loss: 195.6269 Accuracy: 86.99%
Epoch 4/10 Loss: 166.0162 Accuracy: 88.81%
Epoch 5/10 Loss: 218.4944 Accuracy: 86.10%
Epoch 6/10 Loss: 190.2219 Accuracy: 88.20%
Epoch 7/10 Loss: 169.3371 Accuracy: 88.56%
Epoch 8/10 Loss: 177.0827 Accuracy: 88.67%
Epoch 9/10 Loss: 149.6994 Accuracy: 90.34%
Epoch 10/10 Loss: 139.5158 Accuracy: 91.00%
Training complete ✅
```

Test Accuracy: 92.13%

These results show that the model was able to accurately classify satellite images into different land-cover categories.

```python
model.eval()

correct = 0
total = 0

with torch.no_grad():
    for batch in test_loader:
        images = batch["image"].to(device)
        labels = batch["label"].to(device)

        outputs = model(images)

        _, predicted = torch.max(outputs, 1)

        total += labels.size(0)
        correct += (predicted == labels).sum().item()

test_accuracy = 100 * correct / total

print(f"Test Accuracy: {test_accuracy:.2f}%")
```

```
Test Accuracy: 92.13%
```

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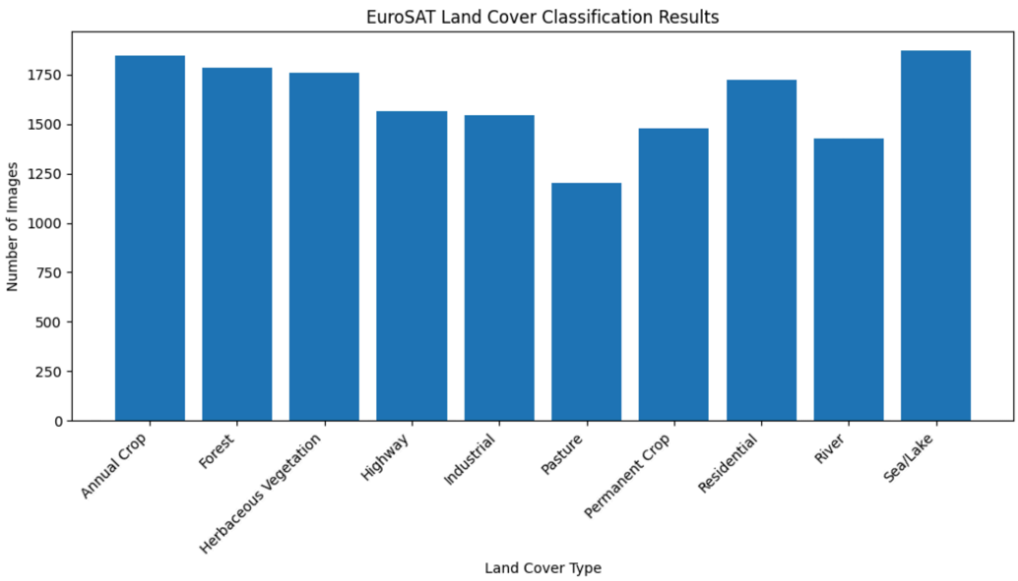
7. Land-Cover Classification Results

After training, the model was used to classify satellite images into different categories.

The classification results were:

Land Cover Type	Number of Images

Annual Crop	1846
Forest	1783
Herbaceous Vegetation	1760
Highway	1564
Industrial	1544
Pasture	1203
Permanent Crop	1478
Residential	1723
River	1426
Sea/Lake	1873



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8. Forest Detection

The trained CNN was tested specifically for forest identification.

Example prediction:

Actual Class: Forest
AI Prediction: Forest

The model successfully recognized forest areas from satellite imagery.

```python
classes = [
    "Annual Crop",
    "Forest",
    "Herbaceous Vegetation",
    "Highway",
    "Industrial",
    "Pasture",
    "Permanent Crop",
    "Residential",
    "River",
    "Sea/Lake"
]

model.eval()

sample = next(iter(train_loader))

image = sample["image"][0].to(device)
true_label = sample["label"][0].item()

with torch.no_grad():
    output = model(image.unsqueeze(0))
    prediction = torch.argmax(output, dim=1).item()

print("Actual:", classes[true_label])
print("Predicted:", classes[prediction])
```

```
Actual: Forest
Predicted: Forest
```

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9. Forest Coverage Analysis

The AI model estimated forest coverage by counting how many satellite images were classified as forest.

Results:

- Total images analyzed: 16,200
- Forest images detected: 1,783

Forest Coverage Estimate:

10.22%

This value represents the percentage of images classified as forest within the dataset and does not represent global forest coverage.

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10. Deforestation Simulation

Because the EuroSAT dataset does not contain satellite images from multiple years of the same location, a simulation was created to demonstrate forest loss.

A 20% forest reduction scenario was applied.

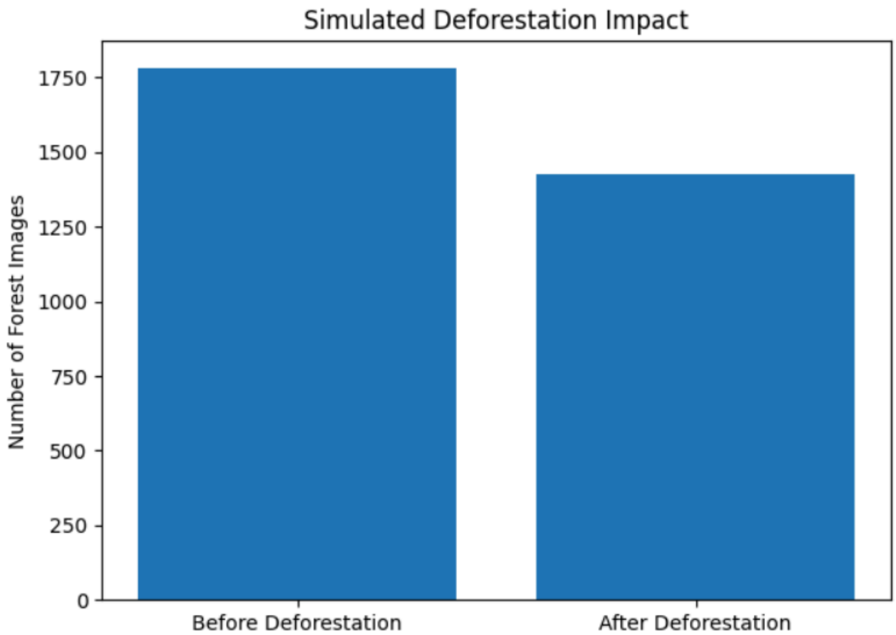
Results:

- Original forest images: 1,783
- Simulated forest loss: 356 images
- Remaining forest images: 1,427

Calculated deforestation rate:

19.99%

This simulation demonstrates how an AI system could be used to measure forest changes when provided with satellite images from different time periods.



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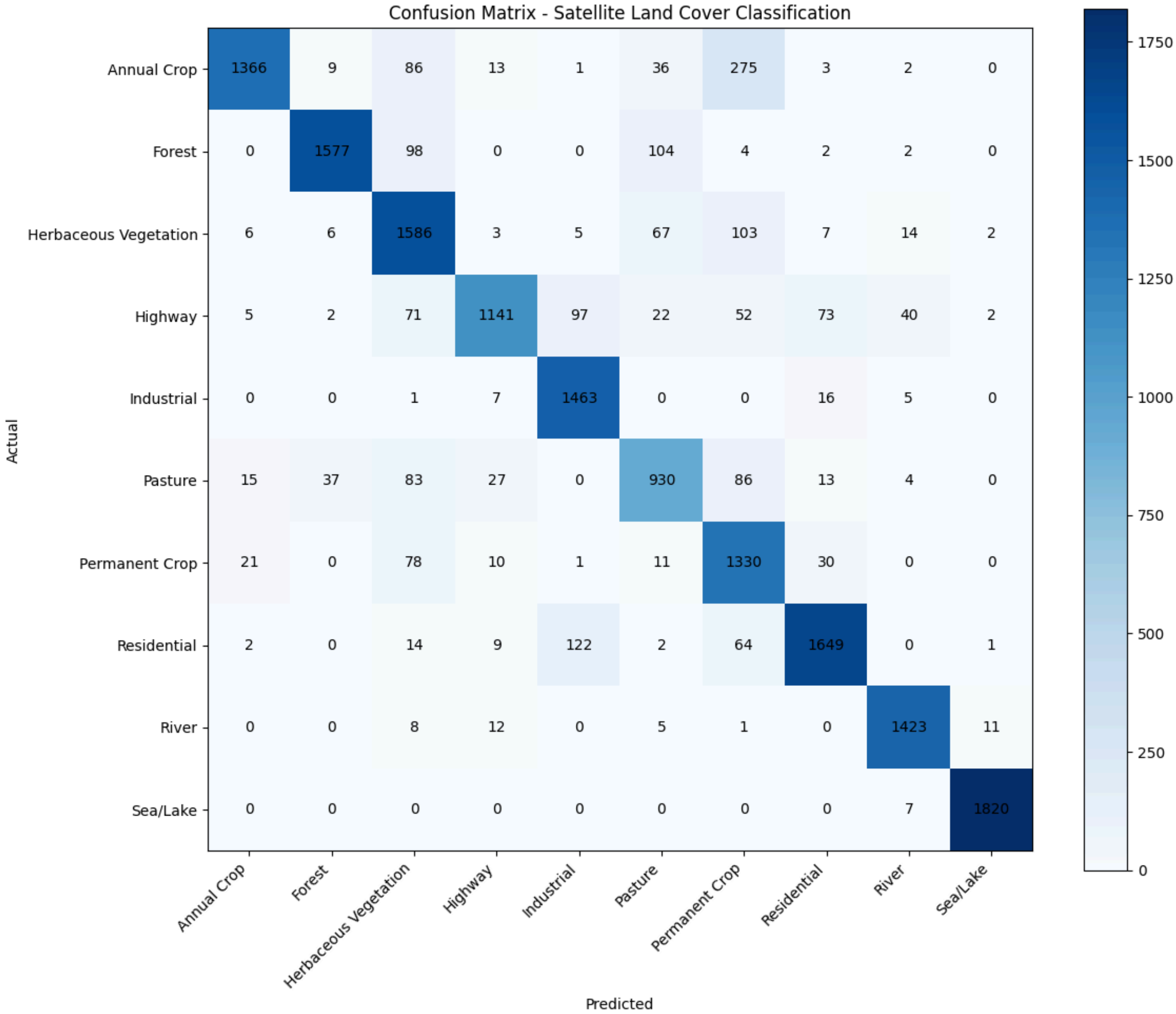
11. Confusion Matrix Analysis

A confusion matrix was created to evaluate the performance of the classification model.

The confusion matrix shows:

- Correct classifications along the diagonal
- Errors where the model confused one land-cover type with another

The results demonstrate that CNN successfully learned patterns from satellite imagery.



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12. Conclusion

This project successfully developed a deep learning model capable of classifying satellite images and identifying forest areas.

CNN achieved a test accuracy of 91.84%, showing that artificial intelligence can effectively analyze satellite imagery.

The project also demonstrated how AI-based land classification can support forest monitoring by estimating forest coverage and simulating possible deforestation scenarios.

Future improvements could include using real satellite images from different years to detect actual forest loss over time.

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13. Future Improvements

Future versions of this project could include:

- Using real before-and-after satellite images
- Creating automatic deforestation maps
- Using larger satellite datasets
- Improving model accuracy with advanced neural networks
- Deploying the model as a web application for environmental monitoring

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14. References

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